



Organic Chemistry: Alcohols

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Alcohols – Introduction

- Alcohols are a family of molecules that contain the **hydroxyl functional group**, -OH
- Their general formula is $\text{C}_n\text{H}_{2n+1}\text{OH}$
- The nomenclature of alcohols follows the pattern **alkan + ol**
- If there are two -OH groups present the molecule is called a **diol**

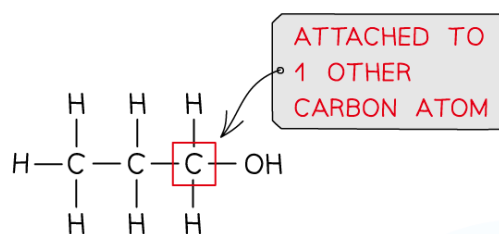
The first four Alcohols and their Structures Table

ALCOHOL	STRUCTURAL FORMULA	DISPLAYED FORMULA
METHANOL	CH_3OH	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{O}-\text{H} \\ \\ \text{H} \end{array}$
ETHANOL	$\text{CH}_3\text{CH}_2\text{OH}$	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}-\text{C}-\text{O}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array}$
PROPANOL	$\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$	$\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{O}-\text{H} \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \end{array}$
BUTANOL	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$	$\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C}-\text{O}-\text{H} \\ \quad \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \end{array}$

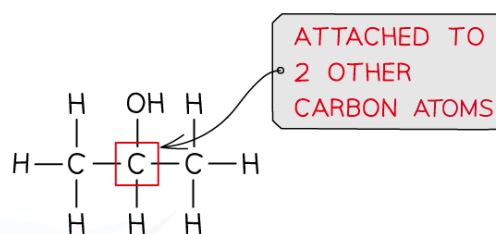
- Alcohols are classified as **primary**, **secondary** or **tertiary** depending on the number of carbons attached to the **functional group** carbon
 - Primary alcohols** are alcohols in which the carbon atom bonded to the -OH group is attached to **one** other carbon atom (or alkyl group)
 - Secondary alcohols** are alcohols in which the carbon atom bonded to the -OH group is attached to **two** other carbon atoms (or alkyl groups)
 - Tertiary alcohols** are alcohols in which the carbon atom bonded to the -OH group is attached to **three** other carbon atoms (or alkyl groups)



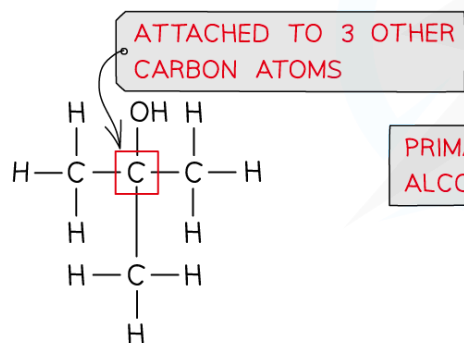
Your notes



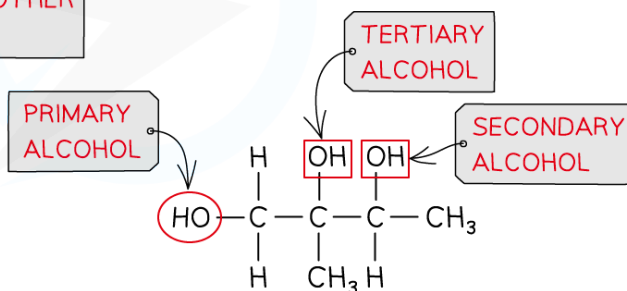
PRIMARY ALCOHOL
PROPAN - 1 - OL



SECONDARY ALCOHOL
PROPAN - 2 - OL



TERTIARY ALCOHOLS
2-METHYL PROPAN - 2 - OL



2-METHYL
BUTAN - 1, 2, 3 - TRIOL

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Classifying primary, secondary and tertiary alcohols and alcohols with more than one alcohol group

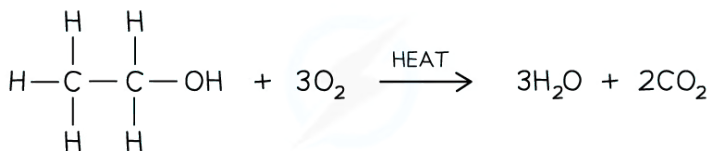


Alcohols – Reactions

Combustion of alcohols

- Alcohols react with oxygen in the air when ignited and undergo **complete combustion** to form carbon dioxide and water

alcohol + oxygen → carbon dioxide + water



Complete combustion of alcohols to produce carbon dioxide and water

Conversions to halogenoalkanes

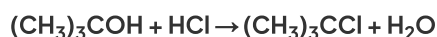
- These reactions involve replacing the hydroxyl group in an alcohol molecule with a halogen atom (known as halogenation)
- Different methods are required for each halogen

Chlorination

- Phosphorus(V) chloride is added to the alcohol resulting in a vigorous reaction at room temperature
 - This means the mixture doesn't need heating
- This reaction is used as a qualitative test for the presence of the -OH group
 - If you add PCl_5 to an unknown liquid the evolution of steamy fumes (HCl gas) is evidence for the presence of the -OH group
- Two inorganic products are formed: phosphoryl chloride and hydrogen chloride



- Chlorination of tertiary alcohols can be carried out in a different way by mixing (shaking) with hydrochloric acid at room temperature
- An example equation for the reaction of 2-methyl propan-2-ol is:



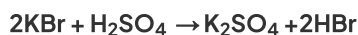
- This reaction does not work for primary and secondary alcohols

Bromination

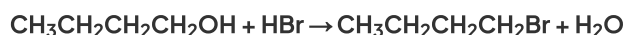


Your notes

- This reaction is carried out using a warmed mixture of potassium bromide and 50% concentrated sulfuric acid with the reacting alcohol
- More concentrated sulfuric acid would oxidise bromide ions to bromine resulting in different products
- The reaction can be written as two equations as the inorganic reactants first react together to form hydrogen bromide and potassium sulfate

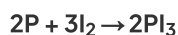


- The resulting hydrogen bromide then reacts with the alcohol, for example the reaction with butan-1-ol would be as follows:

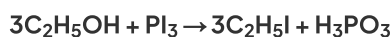


Iodination

- This reaction is carried out using a mixture of red phosphorus and iodine with the alcohol whilst heating under reflux
- Similar to bromination, the reaction can be written as two equations as the inorganic reactants first react to form phosphorus(III) iodide



- The reaction for the iodination of ethanol would be:



- This reaction results in the formation of phosphorous acid as shown above

Dehydration to Alkenes

- Dehydration is done by heating the alcohol with concentrated phosphoric acid
- The reaction is similar to the elimination reaction of a halogenoalkene
 - The OH group and hydrogen of adjacent carbons are removed forming a C=C bond
- The equation for the dehydration of ethanol would be



- Phosphoric acid does not appear in the equation as the water formed dilutes the concentrated phosphoric acid



Alcohols – Oxidation

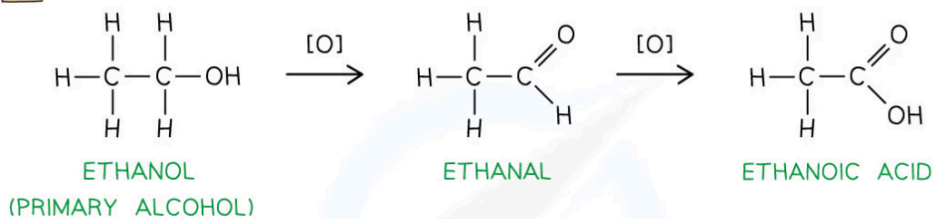
Oxidation of alcohols

- Primary alcohols can be oxidised to form **aldehydes** which can undergo further oxidation to form **carboxylic acids**
- Secondary alcohols can be oxidised to form **ketones** only
- Tertiary alcohols do not undergo oxidation
- The oxidising agents of alcohols include **acidified $K_2Cr_2O_7$**
- **Acidified potassium dichromate(VI)**, $K_2Cr_2O_7$, is an orange oxidising agent
 - Acidified means that the potassium dichromate(VI) is in a solution of **dilute acid** (such as dilute sulfuric acid)
 - For potassium dichromate(VI) to act as an oxidising agent, it itself needs to be reduced
 - This reduction requires hydrogen (H^+) ions which are provided by the acidic medium
 - When alcohols are **oxidised** the orange dichromate ions ($Cr_2O_7^{2-}$) are reduced to green Cr^{3+} ions
- The primary alcohol is added to the oxidising agent and warmed
- The aldehyde product has a lower boiling point than the alcohol reactant so it can be **distilled off** as soon as it forms
- If the aldehyde is not distilled off, further **refluxing** with excess oxidising agent will oxidise it to a carboxylic acid
- Since ketones cannot be further oxidised, the ketone product does not need to be **distilled off** straight away after it has been formed



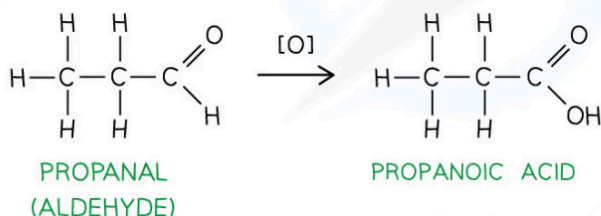
Your notes

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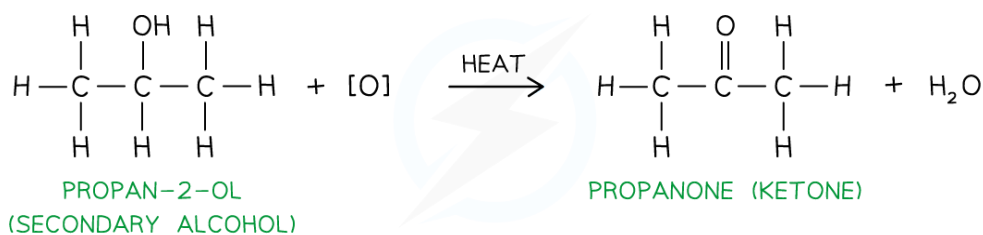
[O] = ACIDIFIED $\text{K}_2\text{Cr}_2\text{O}_7$

2



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Oxidation Stages of Primary Alcohols



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Oxidation of propan-2-ol by acidified $\text{K}_2\text{Cr}_2\text{O}_7$ to form a ketone

- The presence of an aldehyde group ($-\text{CHO}$) in an **unknown compound** can be determined by the **oxidising agents** Fehling's and Tollens' reagents

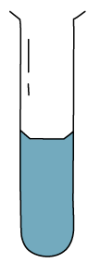
Fehling's solution

- Fehling's solution** is an alkaline solution containing copper(II) ions which act as the oxidising agent
- When **warmed** with an aldehyde, the aldehyde is oxidised to a carboxylic acid and the Cu^{2+} ions are reduced to Cu^+ ions
 - In the alkaline conditions, the carboxylic acid formed will be neutralised to a carboxylate ion (the $-\text{COOH}$ will lose a proton to become $-\text{COO}^-$)
 - The carboxylate ion ($-\text{COO}^-$) will form a salt with a positively charged metal ion such as sodium ($-\text{COO}^-\text{Na}^+$)
- The **clear blue** solution turns **opaque** due to the formation of a **red precipitate**, **copper(I) oxide**

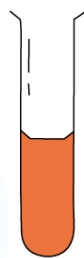
- **Ketones** cannot be oxidised and therefore give a **negative test** when warmed with Fehling's solution



Your notes



CLEAR BLUE FEHLING'S
SOLUTION WITH Cu^{2+} IONS



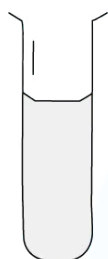
OPAQUE RED PRECIPITATE OF Cu_2O AS:
 - Cu^{2+} IS REDUCED TO Cu^+
 - ALDEHYDE IS OXIDISED TO CARBOXYLIC ACID

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The copper(II) ions in Fehling's solution are oxidising agents, oxidising the aldehyde to a carboxylic acid and getting reduced themselves to copper(I) ions in the Cu_2O precipitate

Tollens' reagent

- **Tollens' reagent** is an aqueous alkaline solution of silver nitrate in excess ammonia solution
 - Tollens' reagent is also called **ammoniacal silver nitrate solution**
- When **warmed** with an aldehyde, the aldehyde is oxidised to a carboxylic acid and the Ag^+ ions are reduced to Ag atoms
 - In the alkaline conditions, the carboxylic acid will become a carboxylate ion and form a salt
- The Ag atoms form a silver 'mirror' on the inside of the tube
- **Ketones** cannot be oxidised and therefore give a **negative test** when warmed with Tollens' reagent



TOLLEN'S SOLUTION
WITH Ag^+ IONS



A SILVER 'MIRROR' OF Ag
ATOMS IS FORMED AS:
 - Ag^+ IS REDUCED TO Ag
 - ALDEHYDE IS OXIDISED
TO CARBOXYLIC ACID

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The Ag^+ ions in Tollens' reagent are oxidising agents, oxidising the aldehyde to a carboxylic acid and getting reduced themselves to silver atoms



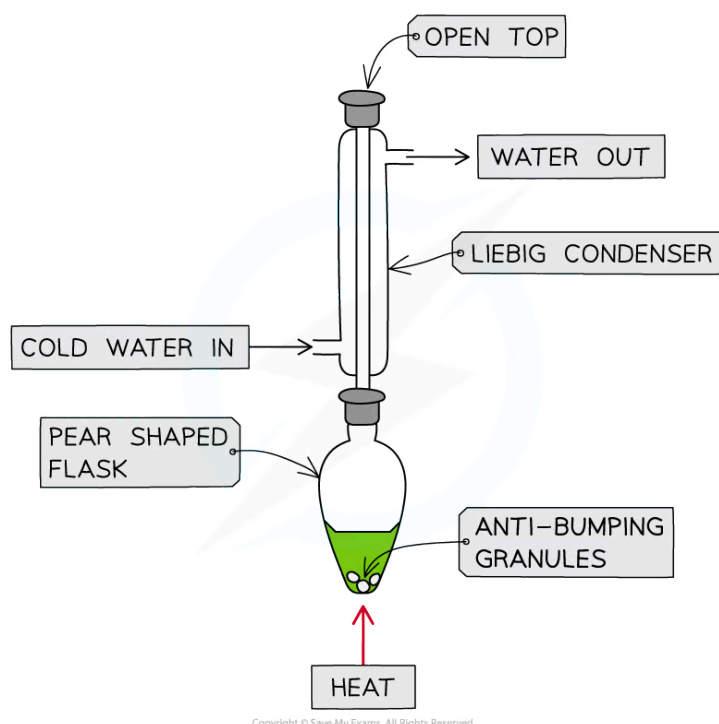
Your notes

Different practical techniques

- Because of the easier oxidation of aldehydes compared to alcohols, two different techniques are used
 - Heating under reflux
 - Distillation with addition

Heating under reflux

- This technique is used when we want full oxidation
 - Producing a carboxylic acid for a primary alcohol
 - Producing a ketone for a secondary alcohol



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Apparatus set up for heating under reflux

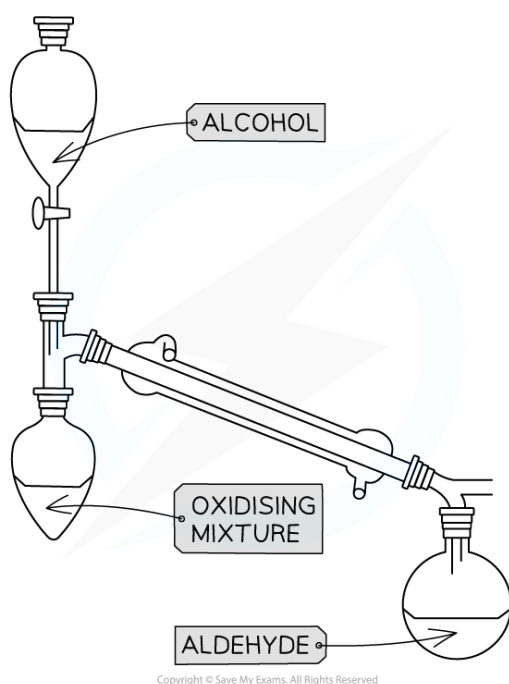
- This set up means any products of oxidation remain in the reaction mixture
- Products which boil off condense in the vertical condenser then return to the heating flask

Distillation with addition

- This technique is used when we do not want to complete oxidation
 - To obtain an aldehyde rather than carboxylic acid for primary alcohol



Your notes



Apparatus set up for distillation with addition

- Only the oxidising agent is heated whilst the alcohol is slowly added
- When the aldehyde is formed it immediately distils off as it has a much lower boiling point than the alcohol used to make it
- The aldehyde is then collected in the receiver